

Department of Physics and Astronomy

First Year

2024-25

Tutorial Sheet 7

PHYS10302 Vibrations and Waves

PHYS10342 Electricity and Magnetism

PHYS10372 Mathematics 2

This sheet covers tutorials in weeks 8 of semester 2.

Problem solving can be a difficult skill to master. If you are finding it tough going, it may be helpful to try the following strategy:

1. Identify what the problem is about. Try to visualise the problem. Drawing a diagram may help.
2. Identify clearly what you are being asked to find.
3. What information have you been given? Sometimes even the simple task of writing down this information can help.
4. What (physics) principles and equations can you use?
5. How do you put points 1, 2, 3 and 4 together to get a solution? What maths will you need?
6. It is usually best to try and solve the problem algebraically, using symbols, even if you are given numerical values. This enables you to check at each stage and at the end, that you have correct and consistent dimensions. The last thing to do is to put numbers in, if a numerical answer is required.
7. Is the answer reasonable? If the answer is numerical, is the order-of-magnitude appropriate? Does your algebraic answer behave in the correct way when the variables take on very large or very small values, or in other limits?

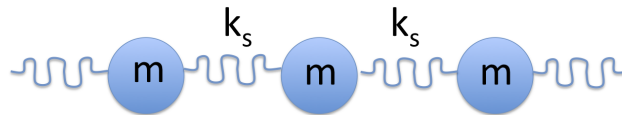
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PHYS10302 Vibrations and Waves

Week 8 Questions

1. (a) An elastic sheet of mass 0.1 kg m^{-2} is stretched under a tension of 50 N m^{-1} . What is the velocity of waves in the sheet?
 (b) A square elastic sheet measures 50 cm by 50 cm . It has a mass of 50 grams . A force of 75 N is applied to each edge. What is the velocity of waves in the sheet?
2. (a) A wave has a wave vector $\mathbf{k} = (1.2, 2.3) \text{ m}^{-1}$. What is the wavelength, and what angle does the direction of wave propagation make with the x -axis?
 (b) A wave in the xy plane has a wavelength of 2.3 mm and is heading at an angle 22° to the x axis and 68° to the y axis. What is its wave vector?
3. Crystalline solids are regular, 3D arrays of atoms. A proper analysis is complicated but some important characteristics of waves in these solids can be deduced from an analysis of the 1D system shown in the figure below. It shows only 3 atoms of an infinite chain of atoms.



The longitudinal motion of the n^{th} atom in this chain is described by

$$m\ddot{x}_n + k_s(2x_n - x_{n+1} - x_{n-1}) = 0,$$

where m is the mass of the atom and k_s is the spring constant. We are not interested in the normal modes of the system, and we just want to know if it can support travelling waves. If it can, for a wave travelling to the right, the motions of adjacent atoms are related to each other by

$$\begin{aligned} x_{n+1}(t) &= x_n(t) \exp(-i\phi), \\ x_{n-1}(t) &= x_n(t) \exp(+i\phi), \end{aligned}$$

where ϕ is the phase lag of the oscillation of the atom with respect to that of the neighbour immediately to the left. ϕ depends on frequency.

- (a) Substitute for x_{n+1} and x_{n-1} in the equation above and show that simple harmonic oscillations of the n^{th} atom are possible at angular frequency ω if

$$\omega = 2 \sin\left(\frac{\phi}{2}\right) \sqrt{\frac{k_s}{m}}.$$

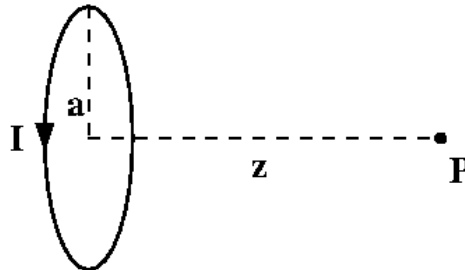
- (b) What is the highest frequency of the wave in such a solid?
- (c) What are the relative motions of adjacent atoms at this maximum frequency?
- (d) Derive an approximate expression for the phase velocity of the waves for the case in which the phase shift ϕ is small and the inter-atomic spacing is d . Hint: how does phase vary as a function of position?

E&M Tutorial Sheet 7: Biot-Savart Law

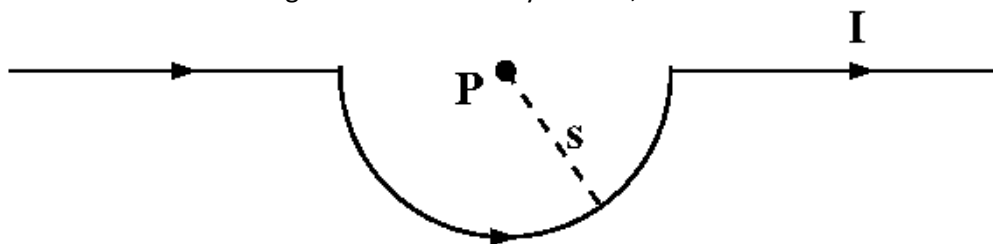
- 1) (Revision) An electron is travelling with velocity $\mathbf{v} = 10^6 \hat{i} + 10^6 \hat{j} \text{ ms}^{-1}$ in the presence of a magnetic field $\mathbf{B} = 0.1 \hat{i} \text{ T}$. Calculate the force on the electron. Without detailed calculation, describe the trajectory of the electron.
- 2) Show that the magnetic field along the axis of a circular loop carrying a current, I , is:

$$\mathbf{B} = \frac{\mu_0 I}{2} \left(\frac{a^2}{(a^2 + z^2)^{3/2}} \right) \hat{z}$$

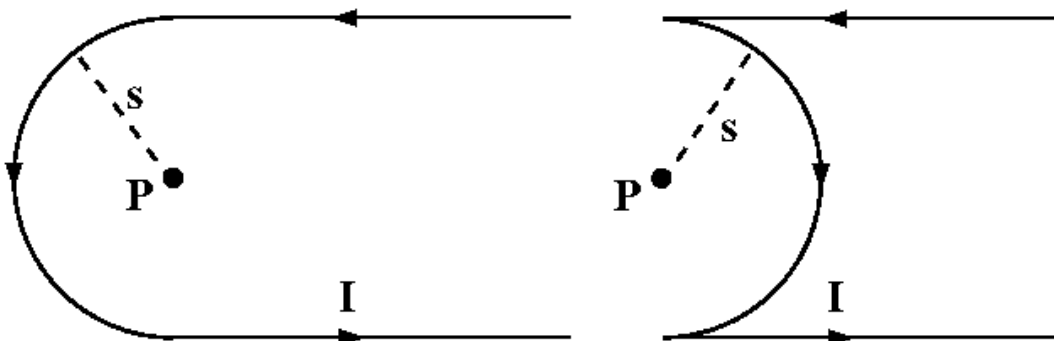
where the other quantities are defined in the diagram below.



- (a) What is the field at $z=0$?
 - (b) What is the field at large values of z ?
 - (c) Sketch a graph of the field strength as a function of z from large negative values to large positive values.
 - (d) How would the formula change if the loop of wire was a coil with N turns of wire?
- 3) The wire shown in the diagram carries a steady current, I .



- (a) What is the contribution to the magnetic field at point P due to the straight parts of the wire?
 - (b) Following the example done in lectures for a full circular loop, calculate the field strength at point P. What direction does the field have at point P?
- 4) Repeat your calculation for the following wires, in which the straight sections are tangential to the semi-circular arcs (and therefore parallel to each other).



PHYS10372 Mathematics 2: Tutorial-07 Questions

These questions cover the material from week-7 of PHYS10372: conservative fields and the Jacobian.

1. Show that the vector field $\mathbf{A} = (-y\hat{\mathbf{i}} + x\hat{\mathbf{j}})(x^2 + y^2)^n$ is a non-conservative field for all values of $n \geq 0$.
2. Show that the vector field $\mathbf{F} = y^2z\hat{\mathbf{i}} + 2xyz\hat{\mathbf{j}} + xy^2\hat{\mathbf{k}}$ is conservative. Determine the potential associated with this field.
3. Calculate the Jacobian when changing from Cartesian to cylindrical polar coordinates.
4. The two-dimensional coordinate system (u, v) is related to the Cartesian coordinates (x, y) through

$$u = x - y$$

$$v = x + y.$$

Calculate the Jacobian, J , such that $dA = dx dy = J du dv$. Then use this to determine the area between the lines defined by $y = x$, $y = -x$, $y = x + 1$ and $y = 1 - x$. Sketch the corresponding areas in the (x, y) and (u, v) coordinate systems.

[Answer: Area = $\frac{1}{2}$]